

Sam Shelly

Weed, CA | (530) 340-1065 | samgshelly@gmail.com | linkedin.com/in/sam-g-shelly | github.com/SamZombie
| samzombie.github.io

SKILLS

Languages: Python, Java, C, C++, C#, OCaml | **ML/AI:** PyTorch, HuggingFace Transformers, LangChain, Ollama, Qdrant | **Tools:** Docker, FastAPI, Streamlit, GitHub Actions, PyMuPDF, RISC-V, pytest

EXPERIENCE

AI Development Intern

Innovative Computing Systems

May 2025 – Jan 2026

Los Angeles, CA

- Hired directly by the COO to lead a Microsoft Copilot Studio Knowledge Agent pilot at a ~60-person company; worked full-time (30 hrs/week) as the primary builder on a 4-person team, meeting weekly with the COO to scope use cases.
- Built Python-based ingestion pipelines to index SharePoint content – SOPs, employee handbook, org hierarchy – for retrieval, and implemented an API integration connecting the agent to the company’s work-ticket system, letting engineers instantly query prior work history on a client.
- Rolled the agent out to a daily-use focus group and began scoping it for a second prospective external client as part of early efforts to commercialize the tool.

Computer Science Dept. Coach

Vassar College

Jan 2024 – May 2026

Poughkeepsie, NY

- Selected for a formal, paid position (8 hrs/week), retained across 4 consecutive semesters; led 3 weekly sessions on algorithms, data structures, and debugging in Java for up to 50 students per semester (200+ total).

PROJECTS

AI Knowledge Base RAG

GitHub

Python, FastAPI, LangChain, Qdrant, Ollama, Mistral 7B, BAAI/bge-large-en-v1.5, PyMuPDF, Docker, Streamlit, pytest, GitHub Actions

Production-style local RAG system for querying government/defense AI policy documents. Designed around free, locally-hosted, open-weight components instead of paid hosted APIs to eliminate cloud dependency and control cost/data exposure.

- Built a custom LLM-as-judge evaluation pipeline achieving 0.95 retrieval recall and 0.92 answer similarity across 10 ground-truth QA pairs, visualized in a Streamlit dashboard.
- Ingests PDFs via PyMuPDF with overlapping chunking; embeds via BAAI/bge-large-en-v1.5 into Qdrant; served via FastAPI REST API, containerized with Docker Compose, with CI/CD via GitHub Actions.

PPO Driving Algorithm

GitHub

Python, stable_baselines3, matplotlib, gymnasium, numpy, pygame

- Trained a 2D autonomous driving agent via Proximal Policy Optimization to navigate a gated course, achieving a 100% gate-pass rate with an asymptotic S-curve reward convergence over ~16M timesteps; demonstrated generalization to unseen course layouts.

EDUCATION

Vassar College

Bachelor of Arts in Computer Science, Minor in Astronomy

Aug 2022 – May 2026

Poughkeepsie, NY

GPA: 3.64/4.00 | Major GPA: 3.68/4.00 | *Coursework:* Data Structures & Algorithms, Analysis of Algorithms, Software Design, Compilers, Theory of Computation, Computer Organization

PUBLICATIONS

Feedback-Driven Magnetic Field Evolution in MW-type Simulated Circumgalactic Medium — KNAC Undergraduate Symposium, 2024 | **Efficacy of NER Models on Fiction and Nonfiction Domains** — CMPU 366 NLP, Vassar College, 2025